Numericals Based on IP

Q1. An IPv4 packet has arrived with the first 8 bits as 01000011, The receiver discards the packet. Why?

Q2. In an IPv4 packet, the value of HLEN is 1011 in binary. How many bytes of options are being carried by this packet?

Q3. In an IPv4 packet, the value of HLEN is 6, and the value of the total length field is 0x0028. How many bytes of data are being carried by this packet?

Q4. In an IPv4 packet, the value of HLEN is 7, and the value of the data bytes are 70. How many bytes are there in total length(in hexadecimal form)?

Q5. An IPv4 packet has arrived with the first few hexadecimal digits as shown.

0x45000028000100000102 . . .

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

Q6. A Datagram of 4000 B (40 B header +3960 B Data) reached at router and must be forwarded to a link with MTU of 600 B. how many fragmentations will be generated and also write MF (more fragment), offset and total length value for all.

Q7. An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. Find the values of the relevant fields in the header of the third IP fragment generated by the router for this packet.

Q8. A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Q9. A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 110. What are the numbers of the first byte and the last byte?

Q10. Calculate the checksum value(in terms of hexadecoimal) of following IPv4 Packet as other values are given as belows.

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4	5	0	28		
1			0	0	
4		17	CHECKSUM		
10.12.14.5					
12.6.7.9					

Q11. An IPv4 packet has arrived with the first 10 bits as 0100010101, The receiver discards the packet. Why?

Q12. In an IPv4 packet, the value of HLEN is 1001 in binary. How many bytes of options are being carried by this packet?

Q13. In an IPv4 packet, the value of HLEN is 8, and the value of the total length field is 0x0030. How many bytes of data are being carried by this packet?

Q14. In an IPv4 packet, the value of HLEN is 8, and the value of the data bytes are 80. How many bytes are there in total length(in hexadecimal form)?

Q15. An IPv4 packet has arrived with the first few hexadecimal digits as shown.

0x76000028000100000204 . . .

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

Q16. A Datagram of 5050 B (40 B header +5010 B Data) reached at router and must be forwarded to a link with MTU of 500 B. how many fragmentations will be generated and also write MF (more fragment), offset and total length value for all.

Q17. An IP router with a Maximum Transmission Unit (MTU) of 1000 bytes has received an IP packet of size 4000 bytes with an IP header of length 20 bytes. Find the values of the relevant fields in the header of the third IP fragment generated by the router for this packet.

Q18. A packet has arrived with an M bit value of 0 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Q19. A packet has arrived in which the offset value is 105, the value of HLEN is 6, and the value of the total length field is 110. What are the numbers of the first byte and the last byte?

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Q20. Calculate the checksum value(in terms of hexadecoimal) of following IPv4 Packet as other values are given as belows.

4	**6**	**0**	**25**	
1			**0**	**0**
3		**17**	**CHECKSUM**	
12.12.15.5				
12.7.7.8				

Numericals based on RSA Algorithm

Q21. Let p=7 and q=13 and plain text message length M=20. Encrypt and Decrypt it using RSA and also calculate its Public Key 'e' and Private Key 'd'.

Q22. Let p=5 and q=17 and plain text message length M=30. Encrypt and Decrypt it using RSA and also calculate its Public Key 'e' and Private Key 'd'.

Q23. Let p=3 and q=5 and plain text message length M=25. Encrypt and Decrypt it using RSA and also calculate its Public Key 'e' and Private Key 'd'.

Numericals based on Data Compression

Q24. Compress the following text using Huffman Encoding technique and calculate total no of bits required for compression.

ELEPHANT IS COOL

Also Calulate their Compression Ratio and Compression Factor.

Q25. Consider the following data stream and compress them using LZ Algorithm and also calculate Total bits required for Data Compression and their Compression ratio

01000000010101111101010

Q26. Consider the following data stream and compress them using Run Length Algorithm and also calculate Total bits required for Data Compression and their Compression ratio

00000000011111111110000000000000111111111111111

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Numericals based on Sliding Window/AIMD

Q27. What will be the size of the window for host A if the value of rwnd is 5000 bytes and the value of cwnd is 7000 bytes?

Q28. What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 6000 bytes and 1110 bytes of received and unprocessed data?

Q29. What will be value of cwnd to be selected as the size of the window for host A if the value of rwnd is 9000 bytes?

Q30. What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 7500 bytes and 1000 bytes of received and unprocessed data?

Q31. In Stop and wait protocol every 4th packet is lost and we need to send total 10 packets so how many transmission it took to send all the packets?

Q32. In GB3 if every 5th packet is lost & we need to send 10 packets so how many retransmissions are required ?

Q33. Consider an instance of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

Numericals based on Network Layer

Q34. Represent the following IP address from dotted decimal notation to hexadecimal notation form.

a) 191.55.76.34

b) 122.46.191.44

c) 244.155.255.255

Q35. Represent the following IP address from hexadecimal notation to decimal notation form.

a) 0x7523F4D2

b) 0x2A76BB43

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c) 0x7389A4C2

Q36. Check the following IP address are valid or not. If valid also identify its Class.

a) 191.55.76.34

b) 122.46.191.44

c) 244.155.255.255

Q37. In Classful addressing, divide a network whose network id is 193.2.5.0 into 4 sub networks and also calculate its host ip range.

Q38. An organization is assigned a class C network address of 222.36.7.0. it uses a net mask of 255.255.255.128 to divide into subnetworks. Which of the following is valid host ip address.

a) 201.35.2.127

b) 201.35.2.135

c) 201.35.2.255

d) None

Q39. In CIDR addressing, divide a network whose network id is 193.2.5.7/24 into 4 sub networks and also calculate its host ip ranges.

Q40. In CIDR addressing under VLSM, divide a network whose network id is 193.2.5.7/24 into 3 sub networks and also calculate its host ip ranges.